

Gradients of the Given Functions

Function a

The gradient of function a is:

$$\partial/\partial x: -1/(x+y) - 2/x$$

$$\partial/\partial y: -1/(x+y) + 1/y$$

$$\partial/\partial z: 0$$

Function b

The gradient of function b is:

$$\partial/\partial x: -12*x/(y^{**2} - z^{**2}) - z*cos(z/(x + z^{**2}))/ (x + z^{**2})^{**2}$$

$$\partial/\partial y: 2*y*(6*x^{**2} + y)/(y^{**2} - z^{**2})^{**2} - 1/(y^{**2} - z^{**2})$$

$$\partial/\partial z: -2*z*(6*x^{**2} + y)/(y^{**2} - z^{**2})^{**2} + (-2*z^{**2}/(x + z^{**2})^{**2} + 1/(x + z^{**2}))*cos(z/(x + z^{**2}))$$

Function c

The gradient of function c is:

$$0 \partial/\partial x: 2*x*z^{**5} + 2*x*z^{**3}$$

$$\partial/\partial y: 3*y^{**2}*z^{**3}$$

$$\partial/\partial z: 5*x^{**2}*z^{**4} + 3*x^{**2}*z^{**2} + 3*y^{**3}*z^{**2}$$

Function d

The gradient of function d is:

$$\partial/\partial x: y/(x+z) + z/(x+y) + \log(y+z)$$

$$\partial/\partial y: x/(y+z) + z/(x+y) + \log(x+z)$$

$$\partial/\partial z: x/(y+z) + y/(x+z) + \log(x+y)$$

Partial Derivatives of Additional Functions

Function a

The partial derivatives of function a are:

$$\partial/\partial x: 4*y^{**3} + \log(y) + y/x$$

$$\partial/\partial y: 12*x*y^{**2} + x/y + z^{**4}*exp(y)*exp(-z^{**4}*exp(y)) + \log(x)$$

$$\partial/\partial z: 4*z^{**3}*exp(y)*exp(-z^{**4}*exp(y))$$

Function b

The partial derivatives of function b are:

$$\partial/\partial x: 3*x^{**2}*\cos(x^{**3} + 2*y^{**2}) - 2*exp(2*x + 4*exp(-y*z))$$

$$\partial/\partial y: 4*y*\cos(x^{**3} + 2*y^{**2}) + 4*z*exp(-y*z)*exp(2*x + 4*exp(-y*z))$$

$$\partial/\partial z: 4*y*exp(-y*z)*exp(2*x + 4*exp(-y*z))$$

Function c

The partial derivatives of function c are:

$$\partial/\partial x: x*y/\sqrt{x^{**2} + y*z}$$

$$\partial/\partial y: y*z/(2*\sqrt{x^{**2} + y*z}) + \sqrt{x^{**2} + y*z}$$

$$\partial/\partial z: y^{**2}/(2*\sqrt{x^{**2} + y*z})$$

Function d

The partial derivatives of function d are:

$$\partial/\partial x: 3*x^{**2}*y^{**3}/(x^{**3} + z^{**2})^{**2} + 2*x/(y^{**2} + 1)$$

$$\partial/\partial y: -2*x^{**2}*y/(y^{**2} + 1)^{**2} - 3*y^{**2}/(x^{**3} + z^{**2})$$

$$\partial/\partial z: 2*y^{**3}*z/(x^{**3} + z^{**2})^{**2}$$

Directional Derivative of the Function

The directional derivative of the function at the given point and direction is:

$$-0.05*\sqrt{3}*pi + 0.4$$